Diego Beltran

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Summary

Software development student with a strong foundation in backend systems, computer vision, and database optimization. Proven track record in delivering scalable applications and contributing to academic research. Recognized for combining technical expertise with problem-solving in real-world projects.

Education

Technological University of Chetumal GPA: 3.9/4.0
Higher Univ. Technician (Jun 2025) & Bachelor's in Software Development (Dec 2026)
Languages: Spanish (Native), English (Conversational)

Experience

Computer Vision Research Assistant *ECOSUR* Apr 2025 – Jul 2025

- Built Python OOP library (OpenCV, NumPy) for area estimation with 99% accuracy in environmental monitoring.
- Refactored 1600+ lines into 6 modular classes, improving maintainability and reusability.
- Designed and executed experiments for marker-based calibration; supported ongoing PhD research.
- Collaborated with a multidisciplinary team to optimize algorithms for large-scale image datasets.

Sojal Reporting System & DevOps *UT Chetumal* Apr 2024 – Apr 2025

- Developed 7 interconnected CRUD systems (users, trucks, routes) with Laravel MVC & Livewire.
- Migrated relational DB to MongoDB, reducing schema complexity by 40% and improving query performance.
- Integrated Mapbox API for real-time route visualization and dynamic route assignment.
- Deployed backend on Ubuntu VM with Apache, implemented monitoring tools for uptime and performance.

Honors & Awards

3rd Place – Best Internship Project Competition Aug 2025
Recognized among 40+ students for outstanding computer vision research and software development.

Selected Coursework

Data Structures & Algorithms, Software Development Methodologies, Operating Systems, Service-Oriented Web Applications, Cloud Databases, Industry 4.0 Web Applications

Technical Skills

Languages: Python, PHP, JavaScript, SQL, HTML/CSS
Frameworks/Tools: Laravel, Livewire, React, Bootstrap, Git, Linux, Apache
Databases: MongoDB, MySQL
Computer Vision: OpenCV, NumPy, Pandas, Matplotlib, cv2.aruco